

Open-Source Security, ICS, & Operations Lab: Installation Guide

This README provides a centralized, standard setup and basic usage guide for deploying your lab environment using Docker.

Prerequisites

Before running any of these tools, ensure you have the following installed on your host machine:

- **Docker Desktop** or **Docker Engine** (v20.10+)
- **Docker Compose** (v2.0+)

Linux Post-Installation: Manage Docker as a Non-Root User

To avoid typing `sudo` before every `docker` command, add your system user to the `docker` group by running the following commands:

```
# Create the docker group if it does not already exist
sudo groupadd docker

# Add your current user to the docker group
sudo usermod -aG docker $USER

# Activate the changes to the groups immediately
newgrp docker
```

Category 1: Mobile Security & OSINT

1. Mobile Security Framework (MobSF)

An automated, all-in-one mobile application (Android/iOS/Windows) pen-testing, malware analysis, and security assessment framework.

- **Installation:**

```
docker run -it --rm -p 8000:8000 opensecurity/mobile-security-framework-mobsf:v4.4.6
```

- **How to Use:**

1. Open your browser and navigate to `http://localhost:8000`.
2. Drag and drop an `.apk` or `.ipa` file into the upload zone.
3. Wait for the automated static analysis report to generate.

2. Drozer

A leading security assessment framework for Android applications, allowing you to assume the role

of an app and interact with the Dalvik VM. (Note: Image repository updated to the active community-maintained repository *drozerdocker/drozer*).

- **Installation & Usage:**

1. To connect to an actual device or emulator, you must install the `drozer-agent.apk` on the target device.
2. Forward the agent port from your host: `adb forward tcp:31415 tcp:31415`.
3. Run the container to connect to the agent:

```
docker run --rm -it --net=host drozerdocker/drozer console connect --server <device_ip>
```

3. PhoneInfoga

One of the most advanced tools to scan international phone numbers using purely open-source intelligence (OSINT).

- **Installation & Basic Scan:**

```
docker run --rm -it sundowndev/phoneinfoga scan -n +15555555555
```

- **Web GUI Mode:**

```
docker run --rm -it -p 5000:5000 sundowndev/phoneinfoga serve -p 5000
```

- *Usage:* Navigate to `http://localhost:5000` to use the interactive dashboard.
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Category 2: Industrial Control Systems (ICS/SCADA)

4. Conpot

An Industrial Control Systems honeypot designed to be easily deployed, modified, and extended to simulate various PLC protocols. (Note: The standalone *honeynet/conpot* Docker Hub image is no longer actively updated; for the latest features, building from the official *mushorg/conpot* GitHub source is recommended).

- **Installation:**

```
docker run -it -p 80:80 -p 102:102 -p 502:502 honeynet/conpot
```

- **How to Use:**

1. The container immediately begins mimicking common ICS protocols like Modbus (502), S7Comm (102), and HTTP (80).
2. Test it from your host using an Nmap scan: `nmap -sV -p 80,102,502 localhost`.

5. ScadaBR / Scada-LTS

An open-source software for developing Supervisory Control and Data Acquisition (SCADA) applications, including Human-Machine Interfaces (HMI). (Note: The original *ScadaBR* project

has evolved into *Scada-LTS*, which provides official Docker support under *scadalts/scadalts*).

- **Installation:**

```
docker run -d -p 8080:8080 --name scadalts scadalts/scadalts:latest
```

- **How to Use:**

1. Navigate to <http://localhost:8080/Scada-LTS>.
2. Log in with the default credentials (typically `admin / admin`).
3. Create data sources (e.g., linking to Conpot or OpenPLC Modbus points) and build custom graphical dashboards.

6. OpenPLC Runtime (v4)

An open-source Programmable Logic Controller implementation that supports the IEC 61131-3 standard for industrial automation.

- **Installation:**

```
# Create persistent volume for runtime data
docker volume create openplc-runtime-data

# Run the official container from GitHub Container Registry
docker run -d --name openplc \
  -p 8443:8443 -p 502:502 \
  --cap-add=SYS_NICE --cap-add=SYS_RESOURCE \
  -v openplc-runtime-data:/var/run/runtime \
  ghcr.io/autonomy-logic/openplc-runtime:latest
```

Note: OpenPLC Runtime v4 uses port 8443 by default for the web GUI (HTTPS) and port 502 for Modbus TCP, meaning it can run alongside Scada-LTS without port conflicts.

- **How to Use:**

1. Open <https://localhost:8443> in your browser (accept the self-signed SSL warning).
2. Log in with `openplc / openplc`.
3. Upload compiled PLC programs (`.st` files) and monitor active registers over Modbus TCP.

Category 3: SIEM, Logs & Threat Intelligence

7. Wazuh (SIEM/EDR)

A free, open-source platform for threat detection, visibility, and compliance tracking across cloud, containers, and endpoints.

- **Installation:**

```
git clone https://github.com/wazuh/wazuh-docker.git
```

```
cd wazuh-docker
git checkout v4.14.6
docker-compose up -d
```

- **How to Use:**

1. Access the dashboard via `https://localhost:5601` (Accept the self-signed SSL warning).
2. Log in with the credentials provided in the repo's environment file (`admin / SecretPassword`).
3. Deploy Wazuh agents to endpoints using the code snippets generated in the UI.

8. Elastic Stack (Logstash / Elasticsearch)

The core infrastructure for searching, analyzing, and visualizing log data in real-time.

- **Installation (Single Node for Testing):**

```
docker run -d --name elasticsearch -p 9200:9200 -p 9300:9300 -e
"discovery.type=single-node" -e "xpack.security.enabled=false"
docker.elastic.co/elasticsearch/elasticsearch:8.19.0
```

- **How to Use:**

1. Verify readiness via `curl http://localhost:9200`.
2. Ship logs to this container using Logstash, Beats, or direct REST API requests.

9. Graylog

A powerful log management system that centralizes data from across your infrastructure.

- **Installation:**

Because Graylog requires MongoDB and OpenSearch/Elasticsearch, a `docker-compose.yml` is highly recommended.

- **How to Use:**

1. Access the web interface at `http://localhost:9000`.
2. Navigate to **System > Inputs** to spin up standard syslog, GELF, or Beats collectors.

10. MISP (Malware Information Sharing Platform)

An open-source threat intelligence platform for sharing, storing, and correlating Indicators of Compromise (IoCs). *(Note: Because MISP relies on multiple services like a database and Redis, deploying via the official `misp-docker` Compose stack is the recommended standard).*

- **Installation:**

```
git clone https://github.com/MISP/misp-docker.git
cd misp-docker
cp template.env .env
docker-compose pull
docker-compose up -d
```

- **How to Use:**

1. Navigate to `https://localhost`.
 2. Log in using default credentials `admin@admin.test / admin`. Change the password immediately.
 3. Subscribe to threat feeds (e.g., AlienVault, CIRCL) to pull active IoCs into your detection pipeline.
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Category 4: Operations & Observability

11. Snipe-IT

A robust, web-based IT asset management system built for tracking hardware, software licenses, and consumables.

- **Installation:**

Snipe-IT requires a database. Use a simple compose configuration:

```
# save as docker-compose.yml and run 'docker-compose up -d'
version: '3'
services:
  snipe-db:
    image: mysql
    environment:
      MYSQL_DATABASE: snipeit
      MYSQL_USER: snipeit
      MYSQL_PASSWORD: password
      MYSQL_ROOT_PASSWORD: root_password
  snipeit:
    image: snipe/snipe-it
    ports:
      - "80:80"
    environment:
      APP_URL: http://localhost
      MYSQL_DATABASE: snipeit
      MYSQL_USER: snipeit
      MYSQL_PASSWORD: password
```

- **How to Use:**

1. Navigate to `http://localhost` to run the initial setup wizard.
2. Create asset tags, categories, and begin checking out hardware to specific users.

12. GlitchTip

An open-source, self-hosted error tracking application that is fully compatible with Sentry client SDKs.

- **Installation:**

```
docker run -d --name glitchtip -p 8000:8000 glitchtip/glitchtip
```

- **How to Use:**

1. Go to `http://localhost:8000` and create an organization/project.
2. Copy the generated **DSN string**.
3. Paste the DSN into your software application code using any standard Sentry SDK to stream errors directly to your dashboard.

13. Grafana

The open-source analytics and interactive visualization web application for time-series metrics.

- **Installation:**

```
docker run -d -p 3000:3000 --name=grafana grafana/grafana:13.1.0
```

- **How to Use:**

1. Navigate to `http://localhost:3000` and sign in (admin / admin).
2. Go to **Connections > Data Sources** and link it to databases like Prometheus, InfluxDB, or your Elasticsearch instance.
3. Create panels using PromQL or Lucene queries to visualize network throughput, security alerts, or log anomalies.

Quick Teardown

To stop and remove running standalone containers from this stack, use:

```
docker stop <container_name_or_id>
docker rm <container_name_or_id>
```

For Compose deployments (like Wazuh or Snipe-IT), navigate to the target directory and run:

```
docker-compose down
```